

PROJECT PROPOSAL & SYNOPSIS

Wipro SDET Capstone Automation Project

Project Title	Enterprise E-Commerce Playwright Automation Framework
Target Application (SUT)	LambdaTest E-Commerce Playground
Application URL	https://ecommerce-playground.lambdatest.io/
Technology Stack	Node.js · Playwright Test · Vanilla JavaScript · Dotenv · Allure Reports · GitHub Actions
Trainee Name	Himanshu Rai
Role / Designation	SDET Trainee — Wipro Technologies
Document Date	May 19, 2026
Total Test Cases	138 test cases across 10 functional modules

#	Module	Test Cases	Type Coverage
01	Authentication	15	Form · Validation · Security · UI
02	Homepage & Navigation	13	Navigation · Performance · UI
03	Product Catalog & Search	15	UI · Interaction
04	Shopping Cart	15	UI · Arithmetic · Validation · Security
05	Wishlist Module	13	UI · Security · Navigation
06	Checkout & Payment	15	Form · Validation · Arithmetic · Navigation
07	User Dashboard	13	Form · Validation · Security · UI
08	Customer Support	13	Form · Validation · UI · Navigation
09	Order History & Management	13	UI · Form · Navigation
10	API Interception & Mocking	13	API · Interception · Security · Performance

2. Objective & Engineering Goals

The primary objective of this Capstone Project is to design, engineer, and deploy a highly resilient, scalable, and ultra-fast test automation framework from scratch. The framework automates end-to-end user journeys on a real-world e-commerce portal, validating all business-critical modules against strict quality benchmarks.

01	Zero-Flakiness Testing Achieve 100% stable execution by enforcing strict auto-waiting and web-first assertions, completely eliminating unreliable hardcoded thread sleeps (setTimeout / page.waitForTimeout).
02	Rapid Parallel Execution Optimize local and cloud run times by enabling fully parallel, multi-threaded test execution using custom worker allocations configured in playwright.config.js.
03	High Reusability — Page Object Model Minimize code duplication by strictly separating web elements (locators) and business actions from test validation scripts using a clean three-layer POM architecture.
04	Automated Cloud Quality via CI/CD Integrate continuous quality gates that automatically trigger the full test suite in GitHub Actions cloud containers on every code push, publishing Allure reports directly.
05	Data-Driven Robustness Ensure complete test independence by utilizing a dynamic test data generator (HelperUtils) to prevent database duplicate record collisions across every test run.

3. Technology Stack Overview

Tool / Technology	Version	Purpose & Framework Role
Playwright Test	v1.44+	Core browser automation engine providing native high-speed auto-waiting and web-first assertions.
Node.js / JavaScript	LTS	Native execution runtime and programming language; matches the language of modern web applications.
Dotenv	v16.4.5	Decouples and secures environment credentials, base URLs, and secrets in local .env config files.
Allure Playwright	v3.0.0	Generates rich, graphic-based HTML test reports, trend dashboards, and suite-level analytics.
GitHub Actions	LTS	Remote CI/CD pipeline that executes the full test suite automatically in Ubuntu cloud containers.

4. Project Folder Structure & Architecture

The project follows a clean, three-layered architecture that completely isolates element locators, page business actions, and test assertion scripts into dedicated folders, ensuring maximum maintainability and zero coupling between automation layers.

PATH	LAYER / PURPOSE
<code>.github/workflows/playwright.yml</code>	CI/CD Layer — GitHub Actions cloud execution pipeline workflow definition
<code>fixtures/pageFixture.js</code>	Fixtures Layer — Custom Playwright fixture injecting Page Objects into test context
<code>pages/HomePage.js</code>	POM Layer — Homepage search, banner interactions, and navigation action methods
<code>pages/LoginPage.js</code>	POM Layer — Login flow, registration form, and profile authentication actions
<code>tests/auth.spec.js</code>	Test Layer — Authentication module assertions and test scenario orchestration
<code>utils/HelperUtils.js</code>	Utilities Layer — Alphanumeric dynamic test data generator to prevent DB collisions
<code>.env</code>	Config — Externalized environment variables: credentials, base URL, secret tokens
<code>playwright.config.js</code>	Config — Master Playwright configuration: browsers, workers, timeouts, reporter
<code>package.json</code>	Config — Project dependencies, version pinning, and custom npm runner scripts

5. Risk Assessment & Flakiness Mitigation

Identified Risk	Impact on Automation	Mitigation Strategy
Flaky / Slow Web Server	Tests timeout due to inconsistent page load delays under load.	Set 15-second navigation timeouts; use auto-retrying assertions polling up to 5 seconds.
Dynamic AJAX Elements	Automation triggers actions before target element is ready.	Playwright natively auto-waits for elements to be stable, visible, and enabled before interaction.
Duplicate Database Records	Registration tests fail as email address already exists in DB.	<code>HelperUtils.getRandomEmail()</code> generates a completely unique email on every single test run.
Brittle CSS / XPath Selectors	HTML layout changes silently break existing locator chains.	Avoid fragile positional XPaths entirely; use stable semantic text and ARIA placeholder locators.

6. 10-Day Implementation Roadmap

Day	Focus Area	Key Deliverables
Day 1	Project Setup & Master Configuration	<code>package.json</code> , <code>playwright.config.js</code> , <code>.env</code> setup, POM placeholders, fixtures, utils, and proposal.
Day 2–3	Authentication & Dashboard Modules	<code>LoginPage</code> + <code>RegisterPage</code> POM classes; 15+ automated scenarios for logins, signups, address book.
Day 4–5	Catalog, Search & Cart Modules	<code>HomePage</code> POM, product detail page, pagination, sorting filters, add-to-cart, and cart calculations.
Day 6–7	Checkout, Payment & Support Modules	Checkout stepper forms, courier selection, credit card mock validation, and Contact Us forms.
Day 8	Order History & Management Modules	Dashboard order history logs, invoice PDF verification, return requests, and cancellation flows.

Day 9	API State Injection & Route Mocking	Pre-auth session state injection via storageState; page.route() for payment and catalog API mocks.
Day 10	Allure Reports, CI/CD Pipeline & Submission	GitHub Actions trigger verification, Allure HTML generation, and final submission packaging.

MODULE 01

AUTHENTICATION

15
TEST
CASES

Validates secure user signup, registration integrity, valid session creation, invalid credential rejection, and secure password recovery mechanisms.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_ATH_001	Successful Account Registration	Fill sign-up form using dynamic generators for random name, unique email, and phone number.	Redirects to success view displaying 'Your Account Has Been Created!'	Form
TC_ATH_002	Duplicate Registration Blocked	Attempt registration using an already registered static email address.	Error alert: 'Warning: E-Mail Address is already registered!'	Validation
TC_ATH_003	Empty First Name Validation	Leave First Name blank; fill all other mandatory registration fields.	Warning: 'First Name must be between 1 and 32 characters!'	Validation
TC_ATH_004	Empty Last Name Validation	Leave Last Name blank; fill all other mandatory registration fields.	Warning: 'Last Name must be between 1 and 32 characters!'	Validation
TC_ATH_005	Empty Telephone Validation	Leave Telephone blank; fill all other mandatory registration fields.	Warning: 'Telephone must be between 3 and 32 characters!'	Validation
TC_ATH_006	Empty Password Validation	Leave Password blank; fill all other mandatory registration fields.	Warning: 'Password must be between 4 and 20 characters!'	Validation
TC_ATH_007	T&C Checkbox Mandate	Fill registration form; leave privacy policy checkbox unchecked.	Banner: 'Warning: You must agree to the Privacy Policy!'	Validation
TC_ATH_008	Successful Login With Valid Credentials	Enter credentials loaded dynamically from secure .env file.	Session starts; navigates directly to the User Dashboard page.	Security
TC_ATH_009	Empty Email Login Validation	Leave Email blank, enter valid password, click login button.	Input field validates locally; blocks API submission.	Validation
TC_ATH_010	Empty Password Login Validation	Enter valid email, leave Password blank, click login button.	Input field validates locally; blocks API submission.	Validation
TC_ATH_011	Invalid Password Error Banner	Enter valid username but incorrect/random password values.	'Warning: No match for E-Mail Address and/or Password.'	Security
TC_ATH_012	Forgot Password Recovery Request	Click forgot password link; enter active email and submit.	'An email with a confirmation link has been sent to your email address.'	UI
TC_ATH_013	Profile Log-Out Terminates Session	Click Log Out in dashboard; attempt to navigate back to account URL.	Session terminates; browser redirects to logout success screen.	Security
TC_ATH_014	Password Mismatch Validation	Type differing values in Password and Confirm Password fields.	'Password confirmation does not match password!'	Validation
TC_ATH_015	Direct Dashboard Access Redirect	Attempt to load Account Dashboard URL in a clean/unauthenticated session.	Browser forces redirect to the Login page for credentials.	Security

Form	Security	UI	Validation	TOTAL
1	4	1	9	15

MODULE 02

HOMEPAGE & NAVIGATION

13
TEST
CASES

Validates catalog left-menu links, category header hover controls, page load performance, responsive asset loading, currency selectors, and visual landing page grids.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_NAV_001	Homepage Loads Within 3 Seconds	Navigate to base URL; record DOMContentLoaded and load execution timestamps.	Page renders fully within 3 seconds without JavaScript console failures.	Performance
TC_NAV_002	Category Header Hover Menus	Hover over primary 'Laptops' category in main navigation bar.	Dropdown displays instantly; clicking items redirects to correct URL categories.	Navigation
TC_NAV_003	Sidebar Left-Menu Category Links	Click categories inside the product catalog side menu (e.g. Components).	Page renders product list; URL paths update with matching parameters.	Navigation
TC_NAV_004	Store Brand Logo Returns To Homepage	Navigate to inner sub-categories; click on primary store header logo.	Browser navigates back to clean home route instantly.	UI
TC_NAV_005	Catalog Grid Pagination Navigation	Click page selector numbers (Page 2 or Next) under list displays.	URL appends page parameters; new product cards load dynamically.	UI
TC_NAV_006	Footer Information Links Resolve	Scroll to footer; click 'About Us', 'Delivery Information', 'Privacy Policy'.	Each link navigates to valid destination page; zero broken links confirmed.	Navigation
TC_NAV_007	Footer Support Section Links Resolve	Scroll to footer; click 'Contact Us', 'Returns', 'Site Map' links.	Each link navigates to valid destination support pages successfully.	Navigation
TC_NAV_008	Header Currency Selector Updates Symbol	Open currency dropdown menu; select 'Euro' or 'US Dollar'.	Entire catalog pricing updates displaying the matching currency sign (€ or \$).	UI
TC_NAV_009	Grid View Toggle Renders Cards	Click the 'Grid View' icon on catalog category listing grid toolbar.	Products rearrange and render as equal-height graphic product cards.	UI
TC_NAV_010	List View Toggle Renders Rows	Click the 'List View' icon on catalog category listing grid toolbar.	Products rearrange and render as horizontal descriptive list rows.	UI
TC_NAV_011	Brand Manufacturer Carousel Loads	Assert manufacturer sliding logo banners at bottom of homepage.	Banners render fully; brand images load without broken asset icons.	UI
TC_NAV_012	Responsive Layout — Tablet 768px	Set browser viewport width to 768px (iPad portrait orientation).	Top navigation collapses to hamburger icon; elements adjust without overlap.	UI
TC_NAV_013	Responsive Layout — Mobile 375px	Set browser viewport width to 375px (iPhone standard width).	Sidebar collapses fully; catalog grid adjusts to single-column card layout.	UI

Navigation	Performance	UI	TOTAL
4	1	8	13

MODULE 03

PRODUCT CATALOG & SEARCH

15
TEST
CASES

Validates global header search functionality, zero-results placeholder messaging, detailed product spec page loading, sort order filters, price range sliders, and brand checkbox filters.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_SRH_001	Global Search Returns Matching Results	Type 'MacBook' or 'HTC' in header search bar and click search.	Renders product grid containing items matching the search keyword.	UI
TC_SRH_002	Invalid Search Shows Empty Message	Type 'InvalidSearchText999' in global search bar and click search.	'There is no product that matches the search criteria.'	UI
TC_SRH_003	Sort By Price Low-To-High	Open 'Sort By' dropdown menu; select 'Price (Low > High)'.	Product listing cards rearranged in ascending order by price tags.	UI
TC_SRH_004	Sort By Price High-To-Low	Open 'Sort By' dropdown menu; select 'Price (High > Low)'.	Product listing cards rearranged in descending order by price tags.	UI
TC_SRH_005	Sort Alphabetically A-Z	Open 'Sort By' dropdown menu; select 'Name (A - Z)'.	Product cards ordered sequentially in ascending alphabetical order.	UI
TC_SRH_006	Sort Alphabetically Z-A	Open 'Sort By' dropdown menu; select 'Name (Z - A)'.	Product cards ordered sequentially in descending alphabetical order.	UI
TC_SRH_007	Brand Manufacturer Checkbox Filter	Check specific manufacturer filter (e.g. Apple) inside sidebar panel.	Grid filters instantly to display only products from the selected brand.	UI
TC_SRH_008	Product Detail Page Loads Specs	Click on a specific product card thumbnail inside a listing.	Product page loads; asserts descriptions, price tags, and large photos.	UI
TC_SRH_009	Empty Search Returns All Products	Clear search input; click search button in header search bar.	Catalog loads displaying the full master product listing.	UI
TC_SRH_010	Search Autocomplete Suggestions	Type 'Mac' in header search bar; wait for autocomplete dropdown.	Suggestion list appears listing MacBook matching titles dynamically.	UI
TC_SRH_011	Price Slider Filters Catalog Range	Drag left and right handlers of the catalog price slider sidebar widget.	List refreshes displaying only products within the custom price boundaries.	UI
TC_SRH_012	Stock Availability Filter Toggle	Check 'Exclude Out of Stock' filter checkbox in sidebar menu.	Listing updates dynamically; displays only active, purchasable products.	UI
TC_SRH_013	Product Comparison Adds To Compare Bar	Click 'Compare Product' icon overlay on 2 different product cards.	Compare bar appears; redirects to side-by-side product spec table.	UI
TC_SRH_014	Category Product Count Badge Matches	Select category (e.g. 'Cameras'); read total count badge displayed.	Badge number matches the total number of product cards listed on the page.	UI
TC_SRH_015	Product Limit Dropdown Changes Grid Size	Open listing display limit dropdown; select '25' products per page.	Catalog display updates successfully; limits items to match the selection.	UI

UI	TOTAL
15	15

MODULE 04

SHOPPING CART

15
TEST
CASES

Validates add-to-cart quick actions, dynamic badge increments, arithmetic calculations for tax and shipping fees, quantity editing, currency toggles, and cart clearing flows.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_CRT_001	Add Single Item To Shopping Cart	Hover on a product card; click 'Add to Cart' quick-action button.	Toast: 'Success: You have added [Product] to your shopping cart!'	UI
TC_CRT_002	Add Multiple Items Updates Badge	Add 3 different items to the cart sequentially via catalog.	Header cart drawer badge dynamically increments to show '3'.	UI
TC_CRT_003	Cart Math — Tax And Shipping	Open Cart; parse base item prices, tax additions, and shipping rate.	Base price + tax + shipping matches grand total displayed value.	Arithmetic
TC_CRT_004	Edit Quantity Recalculates Totals	Open cart; modify item quantity from 1 to 3 and click update.	Product subtotal and grand total adjust with the recalculated price.	Arithmetic
TC_CRT_005	Remove Product Clears Cart Row	Open cart review list; click red Remove/Delete icon on a row.	Table row clears; placeholder text: 'Your shopping cart is empty!'	UI
TC_CRT_006	Zero Quantity Entry Validation	Open Cart; enter 0 in the item quantity field, click update.	System alerts user or automatically removes the item from the list.	Validation
TC_CRT_007	Negative Quantity Entry Blocked	Open Cart; enter -2 in the item quantity field, click update.	Field blocks entry; highlights validation alert or resets to minimum 1.	Validation
TC_CRT_008	Add To Cart From Product Detail Page	Navigate to product detail page; click primary green 'Add to Cart'.	Item added successfully; cart drawer expands dynamically on the right.	UI
TC_CRT_009	Add To Cart From Search Results Grid	Run global search; click cart overlay icon directly from a results card.	Success toast appears; cart drawer reflects the correct item addition.	UI
TC_CRT_010	Cart Drawer Subtotal On Header Hover	Hover over top header cart icon to launch the dropdown cart drawer.	Subtotal displays accurately matching the current cart item values.	UI
TC_CRT_011	Shipping Estimator Updates Total	Enter Country, State, Zip Code inside 'Estimate Shipping & Taxes' widget.	Tax rates apply dynamically; cart totals adjust to match zone fees.	Arithmetic
TC_CRT_012	Promo Coupon Code Lowers Price	Enter valid promo code inside cart review coupon input; click apply.	Subtotal recalculates displaying the correct discount percentage adjustment.	Arithmetic
TC_CRT_013	Checkout Button Redirects To Stepper	Click 'Checkout' action button inside the Cart review panel.	Active session redirects successfully to the secure multi-step Checkout page.	Navigation
TC_CRT_014	Currency Toggle Updates Cart Prices	Open cart with active items; toggle header currency to 'Euro'.	Cart prices and grand total update instantly using new currency conversions.	Arithmetic
TC_CRT_015	Out-Of-Stock Item Checkout Warning	Add unavailable item to cart; click Checkout button to proceed.	Stepper blocks; warning indicates items marked with *** are out of stock.	Security

Arithmetic	Navigation	Security	UI	Validation	TOTAL
5	1	1	6	2	15

MODULE 05

WISHLIST MODULE

13
TEST
CASES

Validates saved items wishlist grids, wishlist badge counts, wishlist item deletions, wishlist-to-cart product transfers, guest access restrictions, and cross-session persistence.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_WSH_001	Save Item To Wishlist Increments Badge	Open a product details page; click 'Add to Wishlist' heart icon.	Success toast displays; header wishlist badge count increases by 1.	UI
TC_WSH_002	Remove Saved Item Updates Wishlist	Open Wishlist dashboard; click red 'Remove' link on a product row.	Item row is deleted; tabular display is refreshed dynamically.	UI
TC_WSH_003	Transfer Wishlist Item To Cart	Open Wishlist page; click blue 'Add to Cart' icon next to a saved item.	Product transfers to cart; cart badge increases, wishlist badge decrements.	UI
TC_WSH_004	Empty Wishlist Shows Placeholder	Remove all items inside the wishlist dashboard page.	Clean placeholder text displays: 'Your wishlist is empty.'	UI
TC_WSH_005	Save Item From Catalog Hover Overlay	Browse catalog grid; click heart icon on product card hover overlay.	Success toast appears; wishlist badge count dynamically updates correctly.	UI
TC_WSH_006	Wishlist Item Pricing Matches Catalog	Verify product price listed inside the wishlist table row.	Pricing matches live category listing prices exactly.	UI
TC_WSH_007	Duplicate Wishlist Addition Alert	Click wishlist heart icon on an already wishlisted product card.	Toast appears alerting user that the product is already in the wishlist.	UI
TC_WSH_008	Wishlist Persists After Logout And Login	Add items to wishlist; log out of profile; log back in fresh.	Open wishlist dashboard; previously added items are accurately saved.	Security
TC_WSH_009	Wishlist Allowed For Out-Of-Stock Items	Browse out-of-stock product details; click 'Add to Wishlist' heart icon.	Item is successfully wishlisted; allows users to track unavailable stock.	UI
TC_WSH_010	Wishlist Thumbnail Redirects To Detail	Open wishlist page; click the visual image thumbnail of a saved item.	Browser redirects to the correct product specification details page.	Navigation
TC_WSH_011	Wishlist Product Compare Button Works	Open Wishlist dashboard; click product comparison icon on a row.	Redirects successfully to product specification comparison tables.	UI
TC_WSH_012	Guest Wishlist Redirect To Login	Attempt to wishlist an item as a guest (logged-out session).	System redirects guest to Login page for dashboard tracking access.	Security
TC_WSH_013	Wishlist Toast Link Opens Dashboard	Click wishlist icon; wait for toast; click 'wishlist' link inside toast.	Navigates directly to the User Wishlist dashboard page.	Navigation

Navigation	Security	UI	TOTAL
2	2	9	13

MODULE 06

CHECKOUT & PAYMENT

15
TEST
CASES

Validates shipping and billing address forms, courier selection logic, mock credit card input format, multi-step checkout stepper navigation, T&C enforcement, and final order confirmation flows.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_CHK_001	Shipping Address Form Validates Input	Navigate to Checkout; fill First Name, Last Name, City, Country, Zone fields.	Form accepts all input; continues to next checkout step without warnings.	Form
TC_CHK_002	Country Dropdown Updates State List	Inside Shipping Address Form; click Country dropdown and select 'India'.	Region/State zone list updates with matching Indian states only.	Form
TC_CHK_003	Use Existing Address Bypasses Form	Select 'I want to use an existing address' radio option in billing section.	Form fields hidden; default profile address is dynamically bound to invoice.	UI
TC_CHK_004	Shipping Courier Selection Adjusts Fees	Select Flat Rate Shipping courier vs. Free Shipping courier option.	Shipping method cost binds to total and adjusts the invoice price correctly.	UI
TC_CHK_005	Billing Comments Text Box Saves Notes	Fill the order comments textarea with custom delivery notes (e.g. 'After 5 PM').	Notes successfully bind to the order data and persist in confirmation.	Form
TC_CHK_006	Mock Credit Card Format Accepted	Type a valid dummy 16-digit credit card number in payment form field.	Form validates format correctly; no invalid input errors are triggered.	Validation
TC_CHK_007	Expired Credit Card Date Validation	Enter an invalid expired date (e.g. 12/20) in the credit card expiry field.	Form validation highlights the field with a red alert warning message.	Validation
TC_CHK_008	Checkout Order Success Page Loads	Click 'Confirm Order' button to execute the purchase transaction.	Navigates to success page displaying: 'Your order has been placed!'	Navigation
TC_CHK_009	Unique Order ID Generated On Success	Parse and extract order identification token on checkout success page.	Displays a unique valid order number (e.g. #10432) for order tracing.	UI
TC_CHK_010	Empty Cart Checkout Navigation Blocked	Attempt to manually load the Checkout URL with an empty cart session.	Redirects session back to clean Home route to prevent a page crash.	Security
TC_CHK_011	Checkout Billing Matches Cart Invoice	Review billing summary listed under the final checkout step.	Product prices, taxes, and shipping exactly match the prior cart totals.	Arithmetic
TC_CHK_012	Stepper Allows Backward Navigation	Transition forward to Step 4; click Step 1 tab to modify shipping address.	Stepper shifts back smoothly; retains all previously entered input values.	UI
TC_CHK_013	Completed Checkout Resets Cart Badge	Execute a complete successful checkout transaction end-to-end.	Cart header badge resets dynamically and displays '0' items.	UI
TC_CHK_014	Unchecked Payment T&C Blocks Order	Attempt checkout confirmation leaving payment terms checkbox unchecked.	Transaction blocks; error: 'Warning: You must agree to the Payment Terms!'	Validation
TC_CHK_015	Return To Cart From Final Checkout Step	Inside the final checkout review step; click the 'View Cart' header link.	Navigates back to Cart list; active transaction item selections remain saved.	Navigation

Arithmetic	Form	Navigation	Security	UI	Validation	TOTAL
1	3	2	1	5	3	15

MODULE 07

USER DASHBOARD

13
TEST
CASES

Validates profile detail modifications, address book record creation and deletion, secondary delivery address updates, password change flows, newsletter subscriptions, and reward point displays.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_DSH_001	Modify Profile First Name Updates Account	Navigate to Edit Account details; change first name and click save.	Success banner: 'Your account has been successfully updated.'	Form
TC_DSH_002	Edit Profile Email Validates Format	Type 'invalidemailformat' inside Edit Profile Email field; click save.	Local email format validation blocks the API submission immediately.	Validation
TC_DSH_003	Address Book Creates New Delivery Entry	Open Address Book panel; click Add New Address, fill details, submit.	New shipping address logs into the user profile list successfully.	Form
TC_DSH_004	Edit Delivery Listing Saves Updates	Select an alternative address in Address Book; click Edit; modify zip code.	Changes successfully save and update the address listing row correctly.	Form
TC_DSH_005	Delete Delivery Listing Updates Records	Open Address Book panel; click 'Delete' button next to an alternative entry.	Address removes instantly from the active address book list.	UI
TC_DSH_006	Password Modification Updates Login	Click Change Password; input a new value; logout; attempt re-login.	System validates new credentials and successfully logs user in.	Security
TC_DSH_007	Dashboard Counters Display Order Counts	Open main dashboard panel view without performing any actions.	Counters accurately display the correct number of past ordered items.	UI
TC_DSH_008	Telephone Field Validates Numeric Format	Open Edit Account; enter letters in the Telephone field; click save.	Submission blocked; prompts user for a valid numeric phone format.	Validation
TC_DSH_009	Newsletter Subscription Checkbox Updates	Check 'Subscribe' checkbox under Newsletter settings section; save.	Dashboard logs user newsletter subscription status as active.	Form
TC_DSH_010	Breadcrumb Navigates Back To Dashboard	Inside inner account settings; click breadcrumb trail link 'Account'.	Smooth redirection back to the user main dashboard overview page.	Navigation
TC_DSH_011	Reward Points Log Displays Statement	Click 'Reward Points' dashboard action panel link.	Loads the statement log containing balances and transaction history.	UI
TC_DSH_012	Downloadable Products List Renders	Click 'Downloads' dashboard action panel link to view purchases.	Tabular list displays valid key codes for purchased virtual asset downloads.	UI
TC_DSH_013	Account Deletion Panel Triggers Caution	Click 'Delete Account' link in the profile dashboard menu section.	Red warning popup alerts user of permanent and irreversible data loss.	Security

Form	Navigation	Security	UI	Validation	TOTAL
4	1	2	4	2	13

MODULE 08

CUSTOMER SUPPORT

13
TEST
CASES

Validates Contact Us page rendering, form submission success flows, mandatory field character limits, empty input validators, email format checks, and success/redirect page navigation.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_SUP_001	Contact Us Page Loads Fully	Navigate to the Contact Us page via navigation or footer link.	Maps, phone dials, and physical store coordinates all render fully.	UI
TC_SUP_002	Submit Contact Us Form Successfully	Navigate to Contact Us; fill Name, Email, and Enquiry (min 10 chars).	Success: 'Your enquiry has been successfully sent to the store owner!'	Form
TC_SUP_003	Empty Email Submission Validation	Fill Name and Enquiry fields; leave Email field completely blank.	Error: 'E-Mail Address does not appear to be valid!'	Validation
TC_SUP_004	Empty Name Submission Validation	Leave Name blank; fill valid Email and complete Enquiry text area.	Error: 'Name must be between 3 and 32 characters!'	Validation
TC_SUP_005	Empty Enquiry Submission Validation	Fill valid Name and Email; leave Enquiry text area completely blank.	Error: 'Enquiry must be between 10 and 3000 characters!'	Validation
TC_SUP_006	Short Enquiry Text Triggers Limit	Type fewer than 10 characters inside the Enquiry text area field.	Error: 'Enquiry must be between 10 and 3000 characters!'	Validation
TC_SUP_007	Excess Enquiry Text Triggers Limit	Type more than 3000 characters inside the Enquiry text area field.	Submission blocked; character limit warning message displayed to user.	Validation
TC_SUP_008	Email Format Validator Blocks Invalid	Type 'missingatsign.com' in the Email input box; click submit button.	Local email format validation blocks the API submission immediately.	Validation
TC_SUP_009	Double Submit Button Block Protects	Click 'Submit' on the contact form multiple times in rapid succession.	System ignores repeated clicks to prevent duplicate email submissions.	UI
TC_SUP_010	Contact Page Print Layout Renders	Press the print action keyboard shortcut inside the Contact Us page.	Renders a printable layout cleanly without navigation bar or footer clutter.	UI
TC_SUP_011	Success Page Continue Button Redirects	Click 'Continue' on the Contact Us form submission success page.	Session redirects smoothly back to the store clean Home homepage route.	Navigation
TC_SUP_012	Site Map Footer Link Launches Page	Scroll to footer section; click the 'Site Map' navigation link.	Navigates to site index page displaying all catalog links and listings.	Navigation
TC_SUP_013	Store Telephone Link Launches Dialer	Scroll to contact section; click the clickable store phone number link.	Launches dialer protocols (tel:+1234567) inside the host OS context.	UI

Form	Navigation	UI	Validation	TOTAL
1	2	4	6	13

MODULE 09

ORDER HISTORY & MANAGEMENT

13
TEST
CASES

Validates Order History list rendering, individual order detail views, PDF invoice generation and printing, product return and cancellation request submissions, and status transition tracking.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_ORD_001	Order History Displays Past Records	Open dashboard profile; click 'Order History' navigation link.	Tabular list displays past order dates, statuses, total costs, and invoice IDs.	UI
TC_ORD_002	Search Order Number Filters Results	Type a specific order number (e.g. 10432) in Order History search.	List filters dynamically displaying only the matched transaction row.	UI
TC_ORD_003	Order Details Shows Purchase Summary	Click 'View Details' button next to a past transaction row.	Detailed receipt view loads showing exact item models, billing address, totals.	UI
TC_ORD_004	Invoice View Launches PDF Print Layout	Open Order Details; click the 'Print/View Invoice' button.	Opens a clean invoice sheet that handles standard PDF printer formats.	UI
TC_ORD_005	Reorder Action Transfers Items To Cart	Open a past order view; click the blue 'Re-order' loop icon button.	Items in past invoice are instantly transferred to the Cart review drawer.	UI
TC_ORD_006	Return Button Redirects To Return Form	Open Order Details; click red 'Return Product(s)' next to an item row.	Redirects to the Product Return Cancellation details request form page.	Navigation
TC_ORD_007	Return Reasons Radio Selector Accepts	In the Return Form, select 'Dead on Arrival' or 'Wrong Item Received'.	Radio selections highlight; cancellation reason registered successfully.	Form
TC_ORD_008	Return Comments Form Submits Request	Fill return reason, type description in return comment box, click submit.	Confirmation page: 'Thank you for submitting your return request!'	Form
TC_ORD_009	Return History Dashboard Tracks Requests	Click 'Return Requests' link inside the dashboard profile section.	Table displays cancellation ticket IDs, current statuses, and details.	UI
TC_ORD_010	Order Cancellation Updates Status Tag	Open an active pending order details page; click 'Cancel Order' button.	Order status updates dynamically in dashboard to display 'Cancelled'.	Form
TC_ORD_011	Checkout Order Shows Processing Status	Complete checkout; navigate immediately to Order History dashboard.	Newly placed checkout order displays active status as 'Processing'.	UI
TC_ORD_012	Past Delivered Order Validates Complete	Assert the status parameter of a past fully delivered transaction row.	Displays final complete status indicator log: 'Complete'.	UI
TC_ORD_013	Invoice Print Launches OS PDF Download	Click the invoice print button inside the browser order detail page.	Launches browser file download dialog or the system printer interface.	UI

Form	Navigation	UI	TOTAL
3	1	9	13

MODULE 10

API INTERCEPTION & MOCKING

13
TEST
CASES

Validates pre-authenticated session injections, Playwright network route interceptions, mock HTTP status codes (500, 429), delayed response simulations, and UI error banner display accuracy.

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_API_001	Pre-Auth Login Injection Speeds Runs	Intercept browser startup; inject valid session storage state from dotenv.	Session opens dynamically; skips Login page; starts directly on Dashboard.	API
TC_API_002	Mock Payment API HTTP 500 Failure	Intercept checkout API endpoint; mock response returning HTTP 500 Error.	Payment stepper stops gracefully; clean user-friendly alert banner displays.	Interception
TC_API_003	Mock Empty Product Catalog Response	Intercept catalog endpoint; mock server response returning empty JSON list.	Browse page displays a clean placeholder message without any crash.	Interception
TC_API_004	Search Suggestion API Interception Mock	Intercept autocomplete search API; return a custom mock suggestions list.	Suggestion dropdown displays the custom mocked list dynamically.	Interception
TC_API_005	HTTP 429 Rate-Limiting Simulation	Route intercept; mock server response returning HTTP 429 Too Many Requests.	UI blocks user inputs; displays slow-down alerts and warning banners.	Interception
TC_API_006	Offline Status Toast Message Mocking	Use route mocking to simulate zero network connection during cart updates.	Toast popup: 'Network Offline - Cart cannot be saved.' displays.	Interception
TC_API_007	Mock Empty Invoice Data Triggers Error	Intercept past order PDF request; return blank values in invoice payload.	PDF generator handles the error gracefully; logs error in dashboard view.	Interception
TC_API_008	Intercept And Inject Security Tokens	Add custom auth headers dynamically during API routing intercepts.	Server authenticates the session seamlessly without UI login registration.	API
TC_API_009	Block Malicious Script Tag Injection	Intercept Profile Update post request; inject <script> tags in payload.	Client validation sanitizes inputs; rejects scripts with format errors.	Security
TC_API_010	Block Third-Party Tracking Scripts	Route intercept; block massive external tracking tags (e.g. analytics).	Overall browser load and test execution times improve dramatically.	Performance
TC_API_011	Mock Slow Database Response Delay	Intercept homepage categories request; delay all responses by 10 seconds.	Page renders clean loading spinners or a reload retry button alert.	Interception
TC_API_012	Mock Dynamic Tax Rate Updates Cart	Intercept checkout tax rates API; mock dynamic adjustments in tax payload.	Cart totals adjust and display values matching the mocked rate additions.	Arithmetic
TC_API_013	Mock Successful Checkout Billing Response	Intercept checkout form submission; mock a success purchase JSON payload.	Bypasses actual banking API delays; forces the checkout success page display.	Interception

API	Arithmetic	Interception	Performance	Security	TOTAL
2	1	8	1	1	13